

IHES - June 2026

Mysterious relations between geometry of Hitchin moduli spaces and VOA modules from 4d $N=2$ theories

Based on joint work with Gukov-Koroteev-Pei-Saberi [\[arXiv:2206.03565\]](#)
& Junkang Huang-Yutai Zhang-Shutong Zhuang [\[arXiv:2412.19647\]](#)

Related works by Peng Shan-Wenbin Yan-Dan Xie, Wenbin Pan-Yiwen Yan

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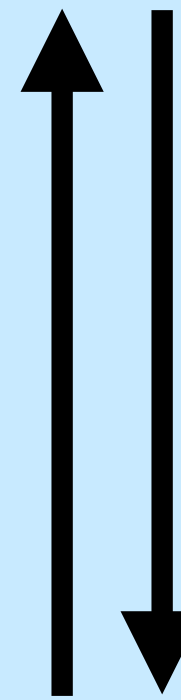
Motivation

4d N=2 Class S theory $\mathcal{T}$ associated to Hitchin system

Higgs branch

assoc. var.

VOA $\mathbb{V}(\mathcal{T})$



chiral
quantization

Hyper-Kähler manifolds



Relation ???

Coulomb branch

Hitchin moduli space

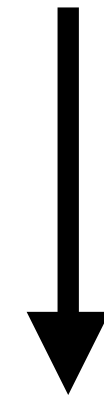
def'n
quantization

Skein algebras
Spherical DAHA



Class S theory

6d (2,0) SCFT of type $\mathfrak{g}$ ADE



$$\mathbb{R}^3 \times S^1 \times C_{g,n}$$

Gaiotto

Gaiotto-Moore-Neitzke

4d N=2 Class S theory $\mathcal{T}_{g,n}$ associated to Hitchin system

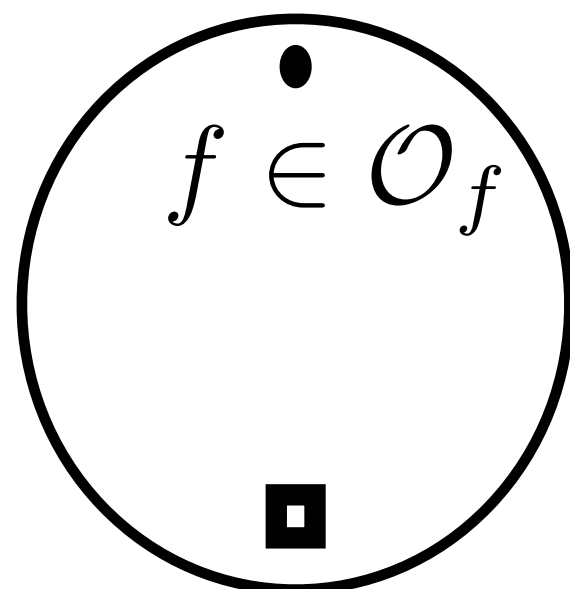
- Regular class S theory ($g \geq 0, n \geq 0$ s.t. $2g - 2 + n > 0$)



$$\phi|_{z_i} \sim \frac{f_i^\vee}{z - z_i} + \dots, \quad f_i \in \mathcal{O}_{f_i}$$

Higgs branch = Star-shaped quiver variety

- Argyres-Douglas type (a sphere with one irregular and one regular puncture)



$$\phi|_0 \sim \frac{f^\vee}{z} + \dots,$$

$$\phi|_\infty \sim \frac{s}{w^{2+\frac{u}{m}}} + \dots, \quad s: \text{semisimple regular}$$

Dan Xie

SECTION 1

4-punctured sphere and brane quantization

4-punctured sphere and Betti moduli space

- Moduli space of flat $\text{SL}(2, \mathbb{C})$ -connections on a four-punctured sphere $C_{0,4}$

$$\mathfrak{X} = \mathcal{M}_{\text{flat}}(C_{0,4}, \text{SL}(2, \mathbb{C})) = \langle M_1, M_2, M_3, M_4 \in \text{SL}(2, \mathbb{C}) \mid M_1 M_2 M_3 M_4 = \text{Id} \rangle / \text{SL}(2, \mathbb{C}) ,$$

- These variables are subject to the **cubic equation**

$$f(x, y, z) = -xyz + x^2 + y^2 + z^2 + \theta_1 x + \theta_2 y + \theta_3 z + \theta_4 = 0 ,$$

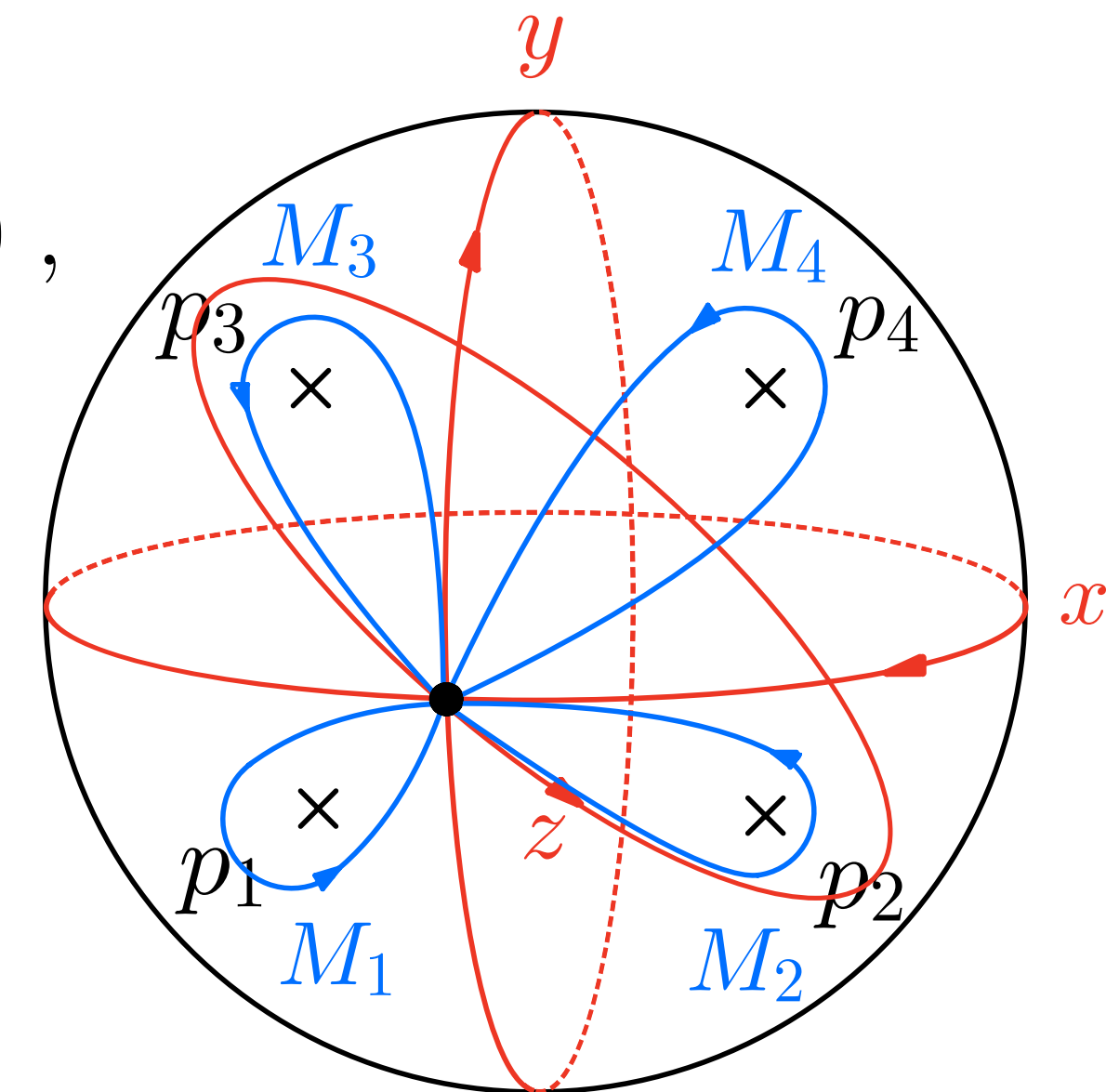
where, using the notation $\bar{f} \equiv f + f^{-1}$, θ_i are expressed by

$$\theta_1 = \chi_{D_4}(\mathbf{8}_V) = \bar{t}_1 \bar{t}_2 + \bar{t}_3 \bar{t}_4 ,$$

$$\theta_2 = \chi_{D_4}(\mathbf{8}_S) = \bar{t}_1 \bar{t}_3 + \bar{t}_2 \bar{t}_4 ,$$

$$\theta_3 = \chi_{D_4}(\mathbf{8}_C) = \bar{t}_1 \bar{t}_4 + \bar{t}_2 \bar{t}_3 ,$$

$$\theta_4 = \chi_{D_4}(\mathbf{adj}) = \bar{t}_1^2 + \bar{t}_2^2 + \bar{t}_3^2 + \bar{t}_4^2 + \bar{t}_1 \bar{t}_2 \bar{t}_3 \bar{t}_4 - 4 .$$



- This space is the Coulomb branch of 4d $\mathcal{N} = 2$ $\text{SU}(2)$ $N_f = 4$ Seiberg-Witten theory on $S^1 \times \mathbb{R}^3$
- Flavor symmetry is $\text{SO}(8)$

Dolbeault moduli space

- Moduli space of parabolic $SU(2)$ -Higgs bundles on a four-punctured sphere $C_{0,4}$

$$\mathcal{M}_H(C_{0,4}, SU(2)) = \{(E, \varphi) \mid E \text{ holom. bdle}, \varphi \in H^0(C_{0,4}, \text{ad}(E) \otimes K)\}_{\text{stab. cond.}},$$

- Tame ramifications $(\alpha_i, \beta_i, \gamma_i) \in (\mathfrak{t}_{A_1})^3$ at punctures $i = (1, 2, 3, 4)$

$$A = \alpha_j d\vartheta + \dots$$

$$\varphi = \frac{1}{2}(\beta_j + i\gamma_j) \frac{dz}{z} + \dots$$

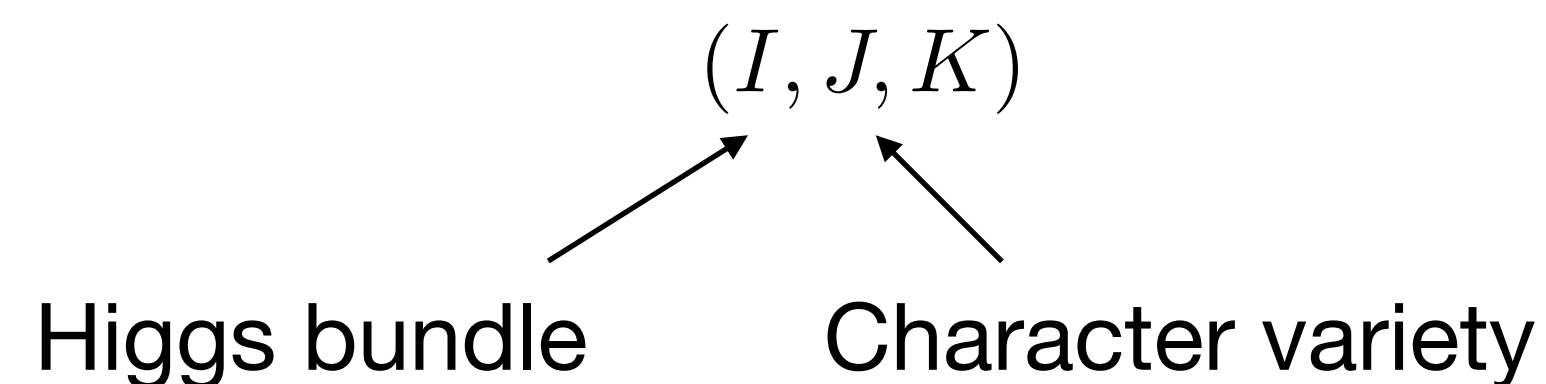
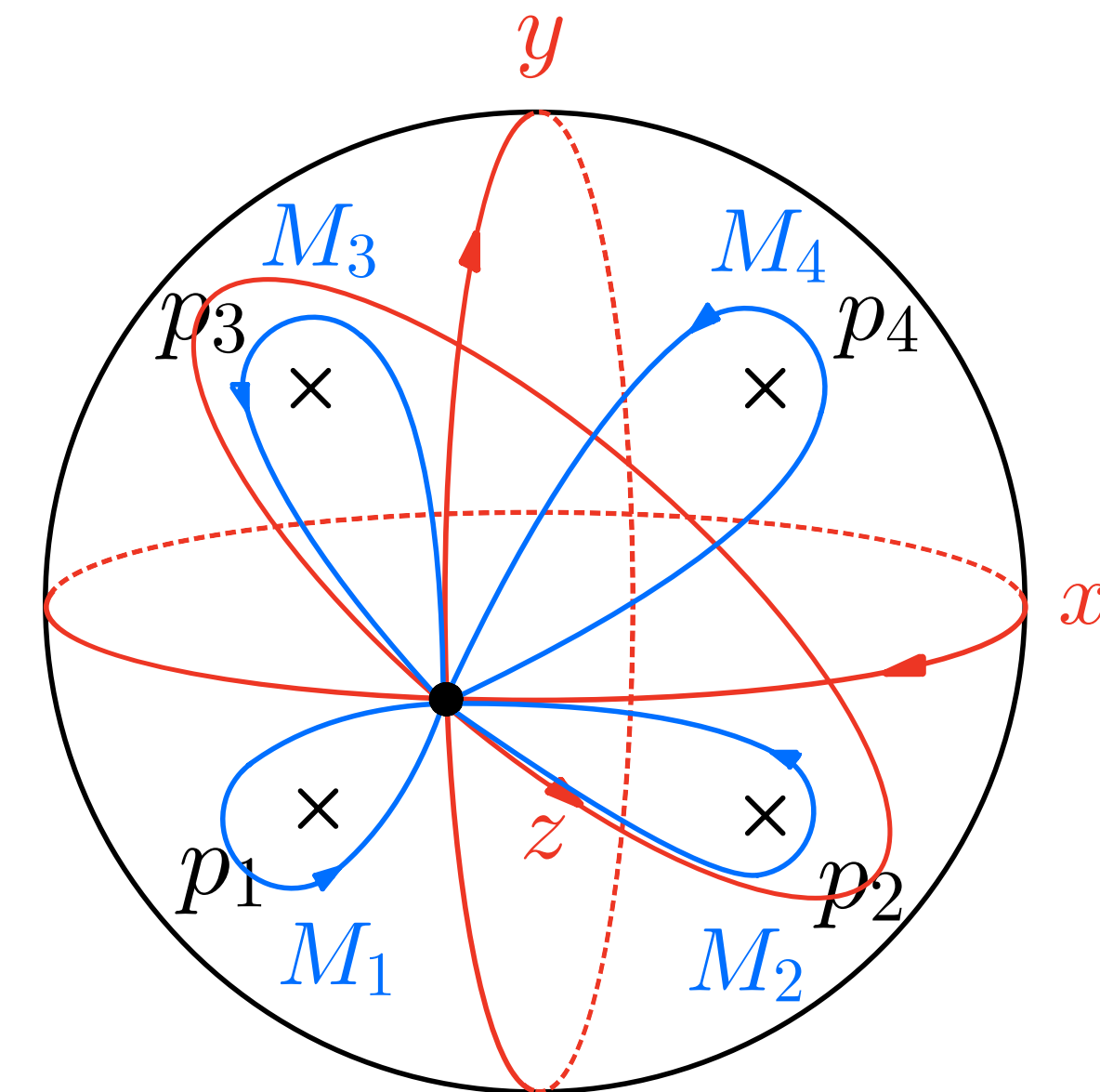
- Ramification data are enhanced to the Cartan subalgebra of D_4

$$\prod_{j=1}^4 \mathfrak{t}_j^{(\alpha)} \times \mathfrak{t}_j^{(\beta)} \times \mathfrak{t}_j^{(\gamma)} \cong \mathfrak{t}_{D_4}^{(\alpha)} \times \mathfrak{t}_{D_4}^{(\beta)} \times \mathfrak{t}_{D_4}^{(\gamma)}$$

- Non-Abelian Hodge correspondence: $\mathfrak{X} = \mathcal{M}_{\text{flat}}(C_{0,4}, SL(2, \mathbb{C})) \cong \mathcal{M}_H(C_{0,4}, SU(2))$

- Complex structures as a hyper-Kähler manifold

$$t_j = \exp(-2\pi(\gamma_j + i\alpha_j)), \quad (j = 1, 2, 3, 4)$$



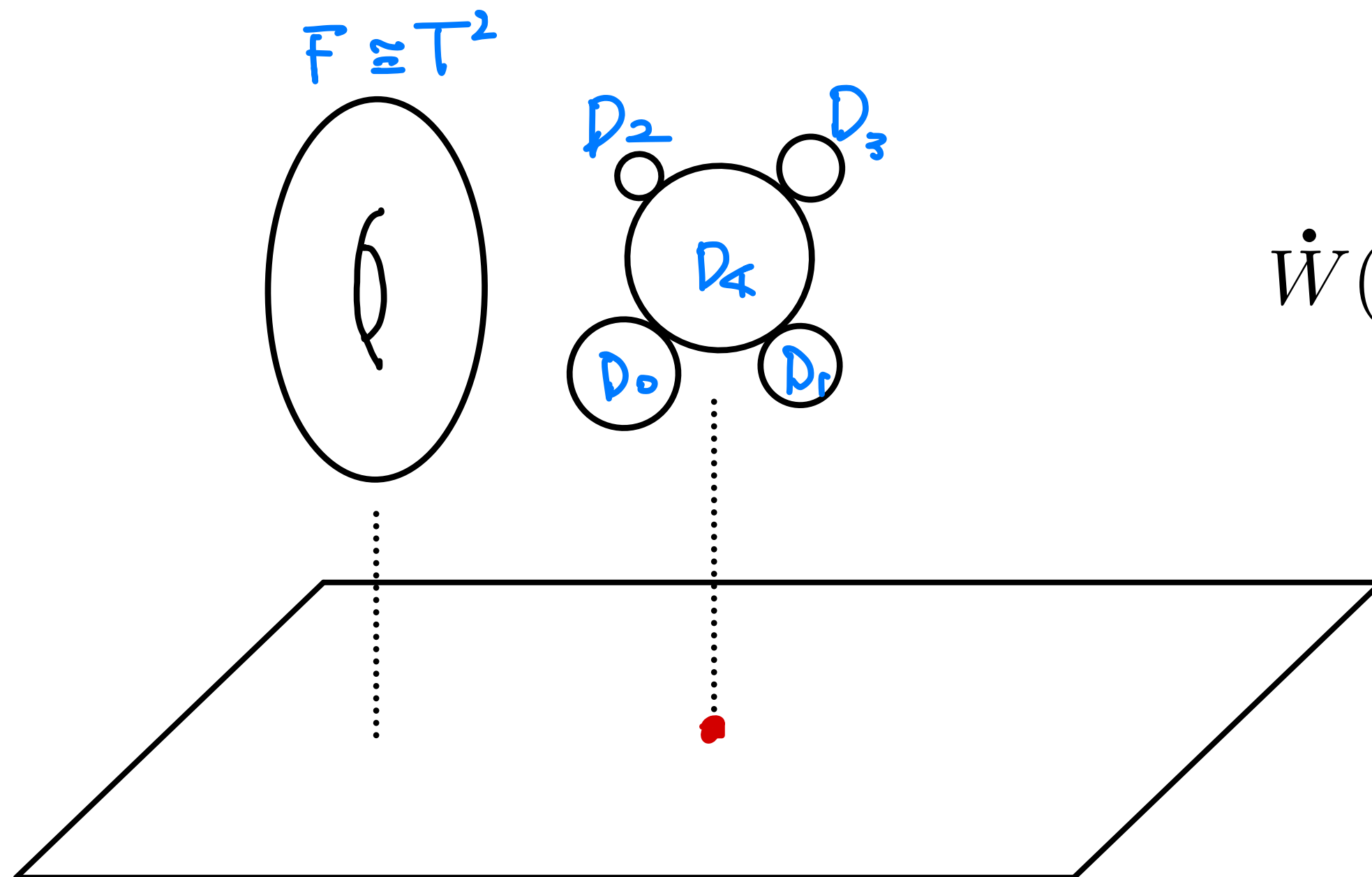
Hitchin fibration

- In complex structure I , the Hitchin moduli space admits the Hitchin fibration over $\mathcal{B}_H$:

$$h : \mathcal{M}_H(C_{0,4}, \mathrm{SU}(2)) \rightarrow \mathcal{B}_H = H^0(C_{0,4}, K_C^{\otimes 2}) .$$

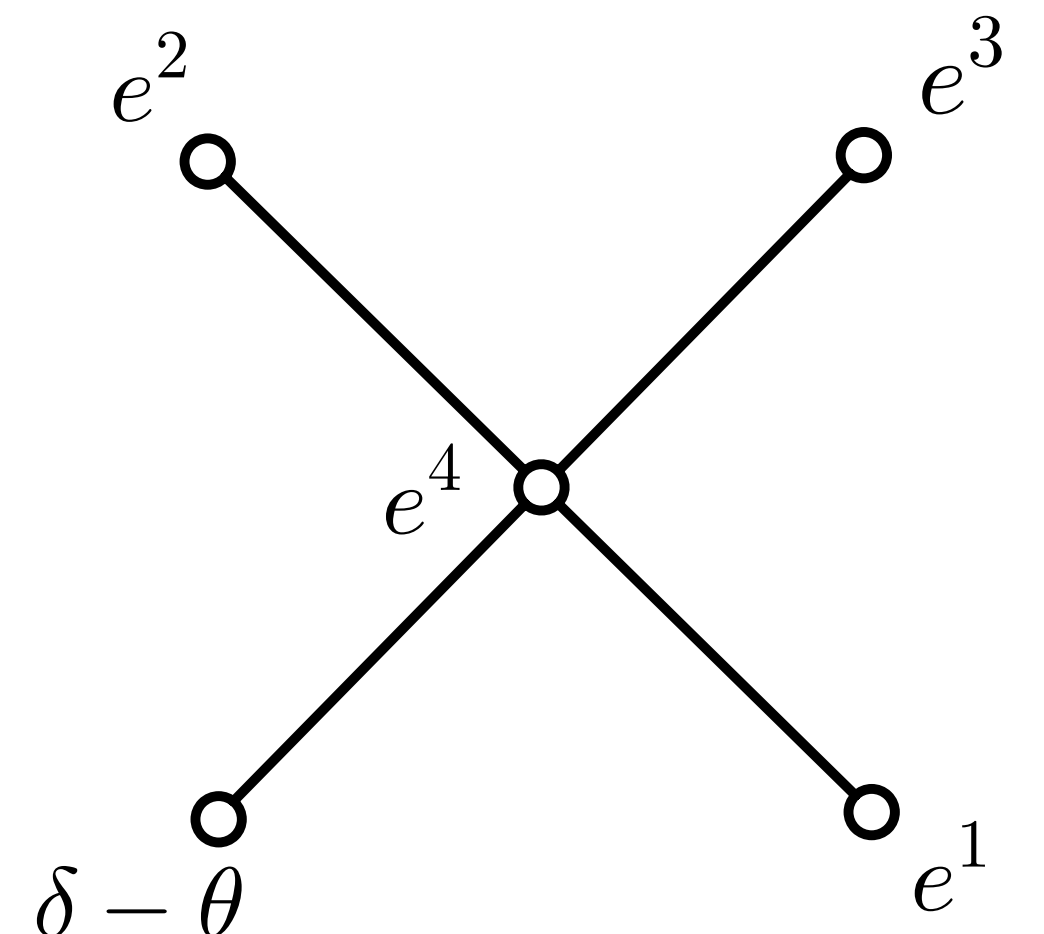
$$(E, \varphi) \mapsto \mathrm{Tr}(\varphi^2)$$

- In our case, the Hitchin fibration is an elliptic fibration with a singular fiber.
- When $\beta_i = 0 = \gamma_i$, the singular fiber is located at the origin.
- fiber class relation: $[\mathbf{F}] = 2[\mathbf{D}_4] + \sum_{i=0}^3 [\mathbf{D}_i]$



$$\dot{W}(D_4) \subset H_2(\mathfrak{X}, \mathbb{Z}) \cong \dot{Q}(D_4) .$$

Afine D_4 Dynkin diagram



Deformation quantization and spherical DAHA

- Deformation quantization of coordinate ring $\mathcal{O}(\mathfrak{X})$ with respect to Ω_J

Oblomkov

$$\mathcal{O}(\mathfrak{X}) \rightsquigarrow \mathcal{O}^q(\mathfrak{X}) \cong S\ddot{H}_{q,t} , \quad [\widehat{f}, \widehat{g}] = -i\hbar \widehat{\{f, g\}} .$$

- It is spherical double affine Hecke algebra $S\ddot{H}_{q,t}$ of type $C^\vee C_1$:

$$\begin{aligned} [x, y]_q &= (q - q^{-1})z + (q^{\frac{1}{2}} - q^{-\frac{1}{2}})\theta_3 , \\ [y, z]_q &= (q - q^{-1})x + (q^{\frac{1}{2}} - q^{-\frac{1}{2}})\theta_1 , \\ [z, x]_q &= (q - q^{-1})y + (q^{\frac{1}{2}} - q^{-\frac{1}{2}})\theta_2 , \\ -q^{\frac{1}{2}}xyz + qx^2 + q^{-1}y^2 + qz^2 + q^{\frac{1}{2}}\theta_1x + q^{-\frac{1}{2}}\theta_2y + q^{\frac{1}{2}}\theta_3z + \theta_4(q) &= 0 , \end{aligned}$$

where, using the notation $\overline{f} \equiv f + f^{-1}$, θ_i are expressed by

$$\begin{aligned} \theta_1 &= \chi_{D_4}(\mathbf{8}_V) = \overline{t_1} \overline{t_2} + \overline{t_3} \overline{t_4} , \\ \theta_2 &= \chi_{D_4}(\mathbf{8}_S) = \overline{t_1} \overline{t_3} + \overline{t_2} \overline{t_4} , \\ \theta_3 &= \chi_{D_4}(\mathbf{8}_C) = \overline{t_1} \overline{t_4} + \overline{t_2} \overline{t_3} , \\ \theta_4(q) &= \overline{t_1}^2 + \overline{t_2}^2 + \overline{t_3}^2 + \overline{t_4}^2 + \overline{t_1} \overline{t_2} \overline{t_3} \overline{t_4} - q - q^{-1} - 2 . \end{aligned}$$

A-model and Brane Quantization

- Brane quantization is geometric approach to representation theory of $\mathcal{O}^q(\mathfrak{X})$ for affine hyper-Kähler manifold $\mathfrak{X}$
- Consider A-model $\Sigma \hookrightarrow (\mathfrak{X}, \omega_{\mathfrak{X}})$
- The distinguished “big” A-brane called “canonical coisotropic brane” $\mathfrak{B}_{cc}$

Gukov-Witten

Kontsevich-Soibelman

$$\mathfrak{B}_{cc} : \begin{array}{c} \mathcal{L} \\ \downarrow \\ \mathfrak{X} \end{array} \quad c_1(\mathcal{L}) = [F/2\pi] \in H^2(\mathfrak{X}, \mathbb{Z}) .$$

where $q = \exp(2\pi i \hbar)$ and

$$\frac{\Omega_J}{2\pi i \hbar} = F + B + i\omega_{\mathfrak{X}}$$

- M is a Lagrangian $\omega_{\mathfrak{X}}|_M = 0$ with respect to $\omega_{\mathfrak{X}}$, so can consider an ordinary A-brane on $\mathfrak{X}$

$$\mathfrak{B}_M : \begin{array}{c} \mathcal{L}' \\ \downarrow \\ M \end{array} \quad \text{local system}$$

Derived equivalence from brane quantization

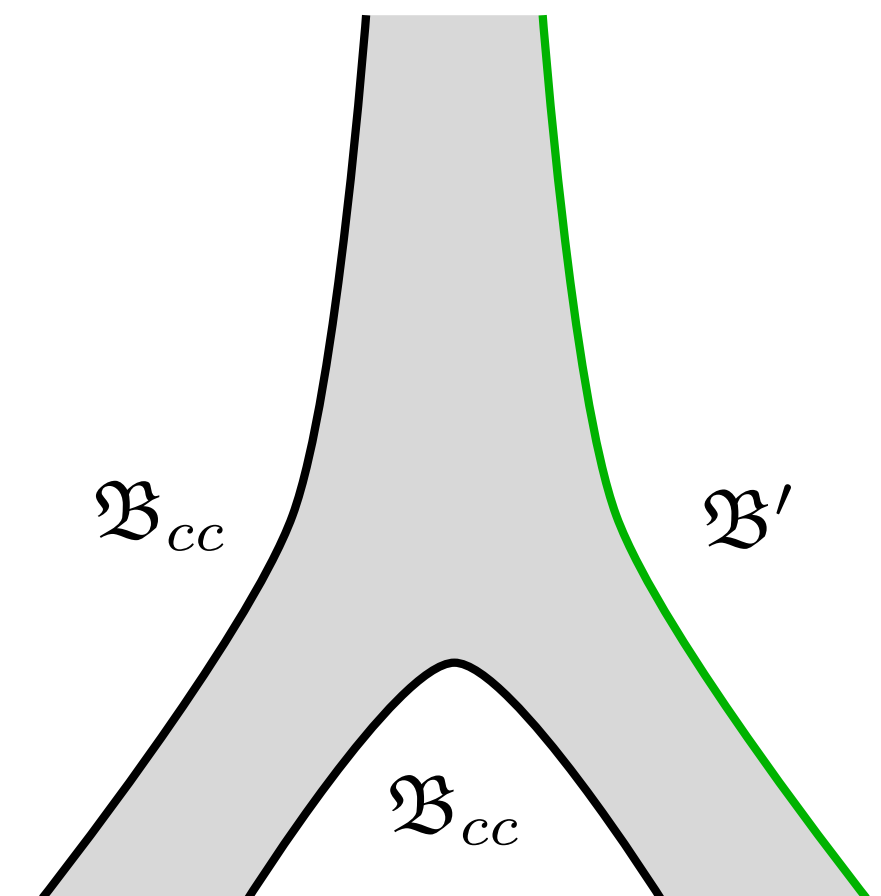
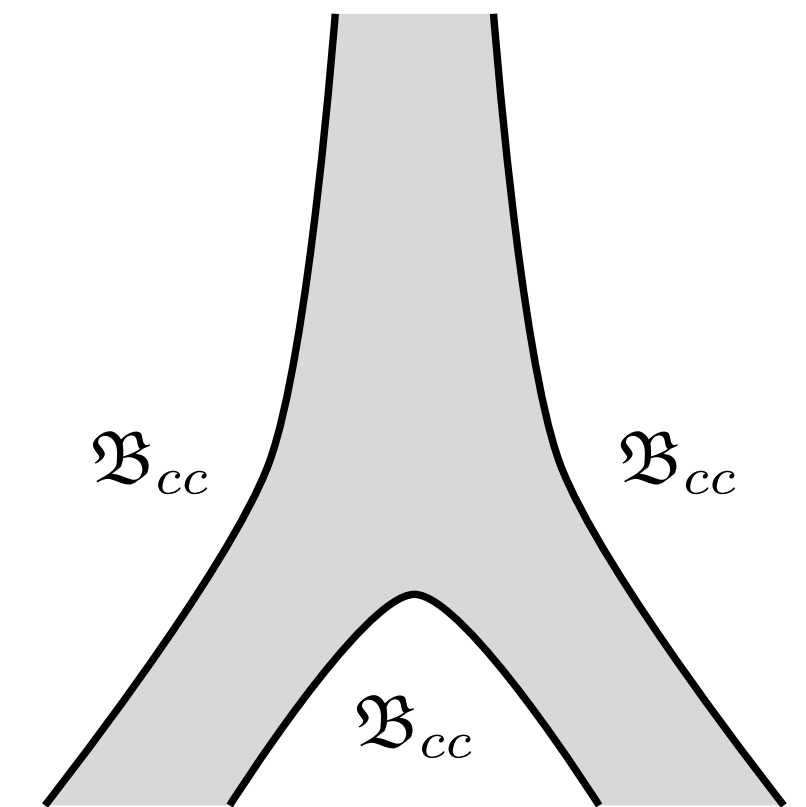
- Joining of strings provides **representation theory** of $\mathcal{O}^q(\mathfrak{X})$ via **A-model** on $\mathfrak{X}$:

$$\begin{aligned}\mathcal{O}^q(\mathfrak{X}) &= \mathrm{Hom}(\mathfrak{B}_{cc}, \mathfrak{B}_{cc}) \\ \bigcirc_Q V_M &= \mathrm{Hom}(\mathfrak{B}_M, \mathfrak{B}_{cc}) .\end{aligned}$$

- This functor is expected to provide derived equivalence

$$\begin{aligned}\mathbf{R}\mathrm{Hom}(-, \mathfrak{B}_{cc}) : D^b A\text{-Brane}(\mathfrak{X}, \omega_{\mathfrak{X}}) &\rightarrow D^b \mathrm{Rep}(S\ddot{H}), \\ \mathfrak{B}_M &\mapsto \mathcal{H} = \mathrm{Hom}(\mathfrak{B}_M, \mathfrak{B}_{cc}) .\end{aligned}$$

Geometric category $\longleftrightarrow$ Representation category



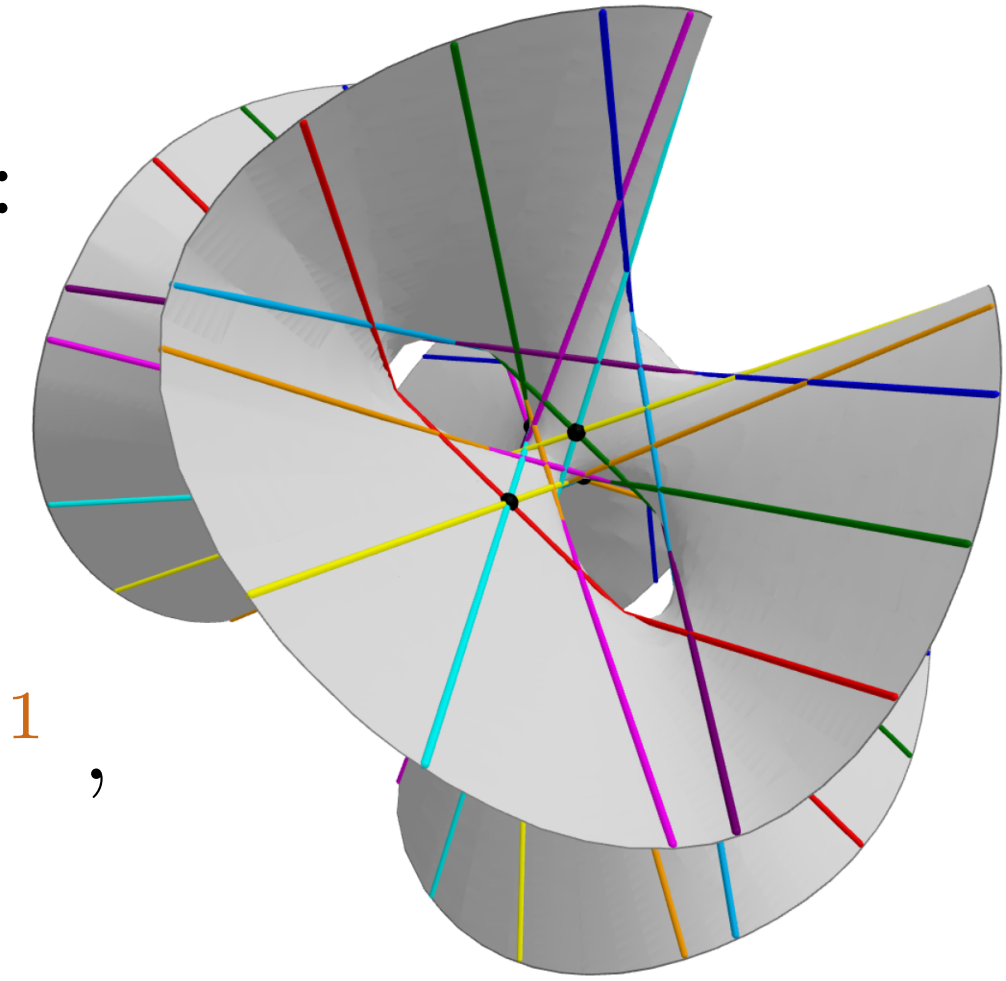
compact A -branes $\longleftrightarrow$ finite-dimensional $\mathcal{O}^q(\mathfrak{X})$ -modules
 non-compact A -branes $\longleftrightarrow$ infinite-dimensional $\mathcal{O}^q(\mathfrak{X})$ -modules

(A,B,A)-branes and polynomial representation

Claim. The (A, B, A) -branes supported on $27 - 3 = 24$ lines in the affine cubic surface $\mathfrak{X}$ correspond to the polynomial representation of $S\ddot{H}_{q,t}$ and its images under certain symmetries.

The polynomial representation acts on the ring of symmetric functions $\mathcal{P} = \mathbb{C}_{q,t}[X^\pm]^{\mathbb{Z}_2}$:

$$\begin{aligned} \text{pol} : S\ddot{H}_{q,t} &\rightarrow \text{End}(\mathcal{P}) , \\ x &\mapsto X + X^{-1} , \\ y &\mapsto A(X; \mathbf{t})(\varpi - 1) + A(X^{-1}; \mathbf{t})(\varpi^{-1} - 1) - q^{\frac{1}{2}} t_1 t_3 - q^{-\frac{1}{2}} (t_1 t_3)^{-1} , \\ z &\mapsto q^{\frac{1}{2}} [X A(X) \varpi + X^{-1} A(X^{-1}) \varpi^{-1}] \\ &\quad - \frac{X + X^{-1}}{q^{\frac{1}{2}} + q^{-\frac{1}{2}}} \left[A(X) + A(X^{-1}) + q^{\frac{1}{2}} t_1 t_3 + q^{-\frac{1}{2}} t_1^{-1} t_3^{-1} \right] - \frac{\theta_3}{q^{\frac{1}{2}} + q^{-\frac{1}{2}}} . \end{aligned}$$



Askey-Wilson polynomial is the eigenfunction of y under polynomial representation

$$\text{pol}(y) \cdot P_n(x; q, \mathbf{t}) = (-q^{n+\frac{1}{2}} t_1 t_3 - q^{-n-\frac{1}{2}} t_1^{-1} t_3^{-1}) P_n(x; q, \mathbf{t})$$

This corresponds to an (A, B, A) -branes supported on

$$\mathbf{P}_{(1,0,1,0)} = \{y = -t_1 t_3 - t_1^{-1} t_3^{-1}, z = -t_1 t_3 x - t_1 t_4 - t_1 t_4^{-1} - t_2 t_3 - t_2^{-1} t_3\}$$

Compact A-branes and finite-dimensional modules

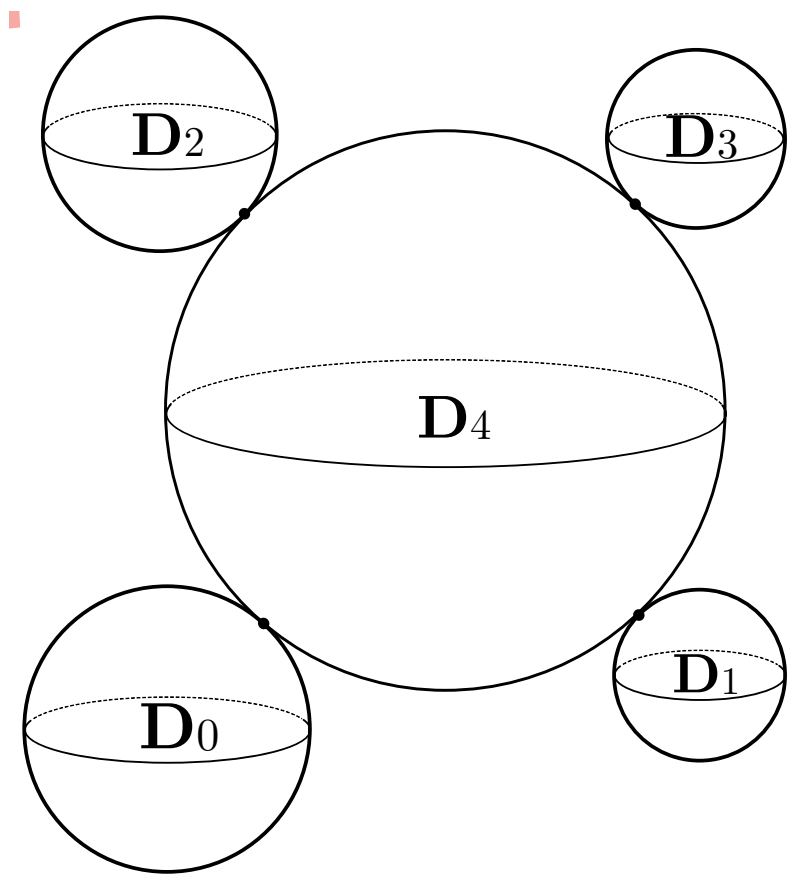
Irreducible components at a Hitchin fiber are (B, A, A) -branes which is Lagrangian when $\hbar \in \mathbb{R}$

$$\ell_i = \dim \mathcal{D}_i = \int_{\mathbf{D}_i} \frac{\Omega_J}{2\pi i \hbar} = \frac{\alpha \cdot e^i}{\hbar} \, .$$

Claim. The functor $\mathrm{Hom}(-, \mathfrak{B}_{cc})$ restricts to a derived equivalence of the full subcategory of compact Lagrangian A -branes of $\mathfrak{X}$ and the category of finite-dimensional $S\ddot{H}_{q,t}$ -modules.

finite-dim rep	shortening condition	A -brane	A -brane condition
$\mathcal{F}_m^{(x_m)}$	$q^m = 1$	$\mathfrak{B}_{\mathbf{F}}^{(x_m)}$	$m = \frac{1}{\hbar}$
$\mathcal{V}_k$	$q^k = t_1^{-2}$	$\mathfrak{B}_{\mathbf{V}}$	$k = \frac{2\alpha_1}{\hbar}$
$\mathcal{D}_{\ell_1}^{(1)}$	$q^{\ell_1} = t_1 t_2 t_3 t_4$	$\mathfrak{B}_{\mathbf{D}_1}$	$\ell_1 = \frac{1}{\hbar} - \frac{\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4}{\hbar}$
$\mathcal{D}_{\ell_2}^{(2)}$	$q^{\ell_2} = t_1 t_2 t_3^{-1} t_4^{-1}$	$\mathfrak{B}_{\mathbf{D}_2}$	$\ell_2 = \frac{-\alpha_1 - \alpha_2 + \alpha_3 + \alpha_4}{\hbar}$
$\mathcal{D}_{\ell_3}^{(3)}$	$q^{\ell_3} = t_1 t_2^{-1} t_3 t_4^{-1}$	$\mathfrak{B}_{\mathbf{D}_3}$	$\ell_3 = \frac{-\alpha_1 + \alpha_2 - \alpha_3 + \alpha_4}{\hbar}$
$\mathcal{D}_{\ell_4}^{(4)}$	$q^{\ell_4} = t_1 t_2^{-1} t_3^{-1} t_4$	$\mathfrak{B}_{\mathbf{D}_4}$	$\ell_4 = \frac{-\alpha_1 + \alpha_2 + \alpha_3 - \alpha_4}{\hbar}$

Compact A-branes and finite-dimensional modules

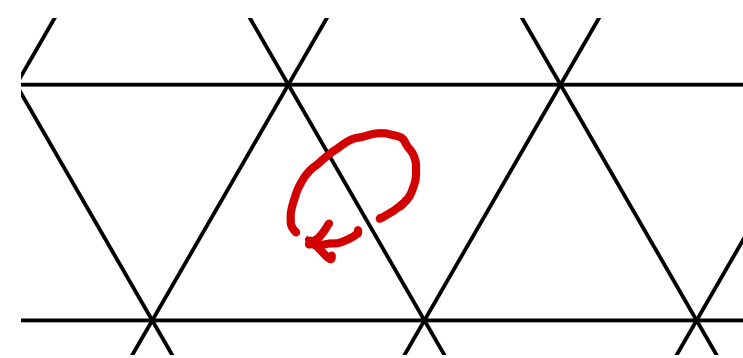


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$\mathcal{D}_{\ell_4}^{(4)}$	$q^{\ell_4} = t_1 t_2^{-1} t_3^{-1} t_4$	$\mathfrak{B}_{\mathbf{D}_4}$	$\ell_4 = \frac{-\alpha_1 + \alpha_2 + \alpha_3 - \alpha_4}{\hbar}$

Affine braid group action on categories

- The A -model depends only on complexified Kähler moduli, but is independent of complex structure.
- If $\hbar \in \mathbb{R}$, then (α, β) is the complex structure, while $\gamma + i\eta$ is the complexified Kähler moduli.
- Variations along a loop in the complex structure moduli space (avoiding singularities) induce non-trivial transformations of the A -brane category.



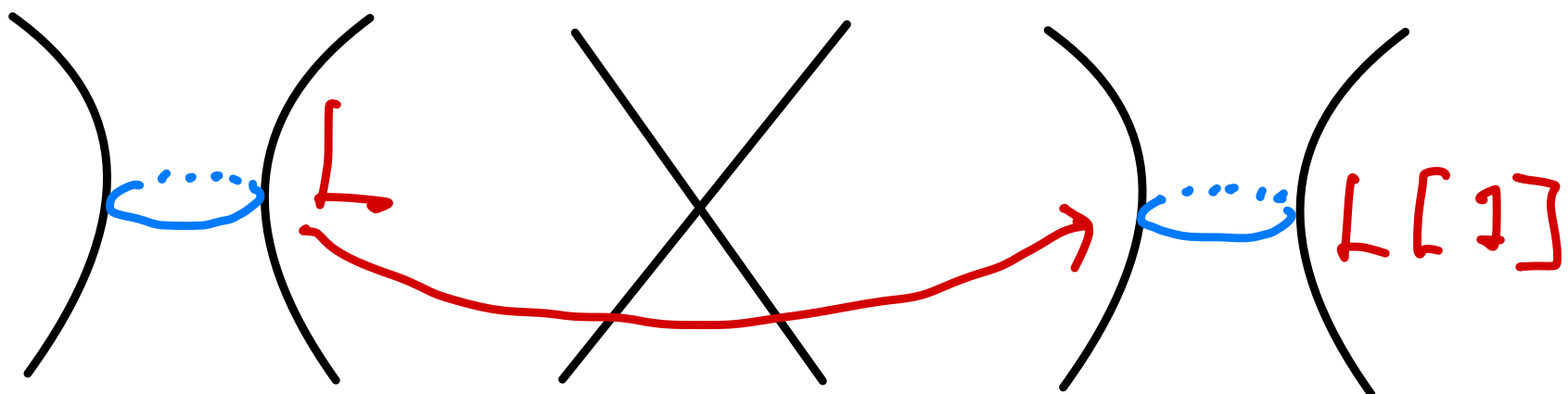
P. Seidel

- When $\gamma + i\eta = 0$, the affine braid group $\dot{\text{Br}}(D_4)$ acts on the category:

$$\begin{array}{ccc} & \dot{\text{Br}}(D_4) & \\ \circlearrowleft & & \circlearrowright \\ D^b A\text{-Brane}(\mathfrak{X}, \omega_{\mathfrak{X}})_{\gamma=0=\eta} & \cong & D^b \text{Rep}(S\ddot{H}_{q,t})_{|q|=1=|t_j|} \end{array}$$

where $T_a([\mathbf{D}_b]) = [\mathbf{D}_b] + ([\mathbf{D}_a] \cdot [\mathbf{D}_b])[\mathbf{D}_a]$ are subject to

$$\begin{aligned} T_j T_k &= T_k T_j \\ T_j T_4 T_j &= T_4 T_j T_4 \end{aligned}$$



SECTION 2

VOA modules vs $\mathbb{C}^\times$ -fixed components of $\mathcal{M}_H$

SCFT/VOA Correspondence

Beem-Lemos-Liendo-Peelaers-Rastelli-van Rees,

$$4d \mathcal{N} = 2 \text{ SCFT } \mathcal{T} \longrightarrow \text{VOA } \mathbb{V}(\mathcal{T})$$

- The 2d central charge is related to the 4d central charge by

$$c_{2d} = -12c_{4d}$$

- The vacuum character of the VOA equals the Schur index of the 4d theory:

$$\chi_{\mathbb{V}(\mathcal{T})} = I_{\text{Schur}}(\mathcal{T})$$

- The flavor symmetry algebra decomposes into simple factors

$$\mathfrak{g}_F = \bigoplus_i \mathfrak{g}_i \oplus \mathfrak{u}(1)^{\oplus s},$$

and each simple factor gives an affine Lie subalgebra $L_{k^{2d}}(\mathfrak{g}_i) \subset \mathbb{V}(\mathcal{T})$, with negative level

$$k_i^{2d} = -k_i^{4d}$$

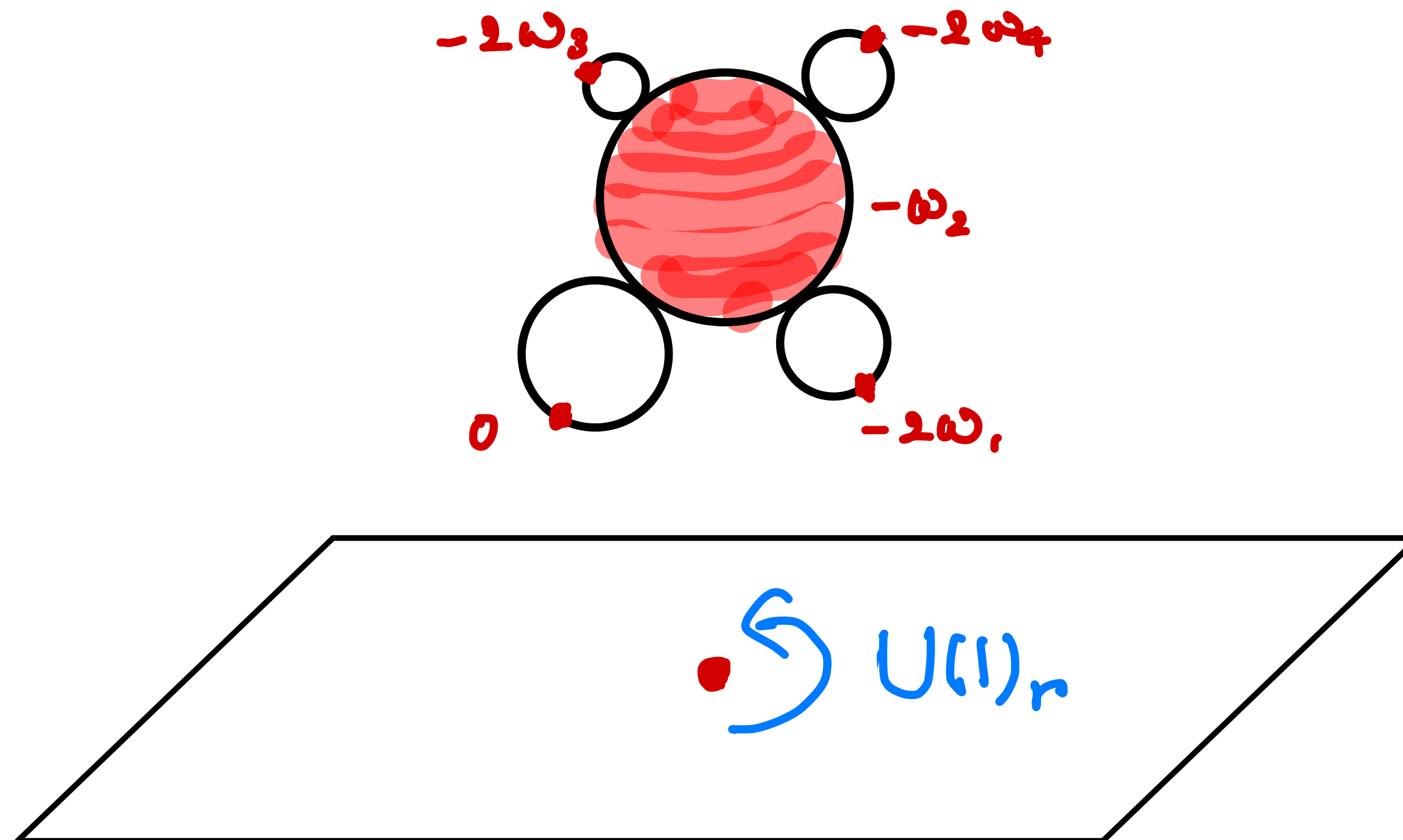
- The associated variety $X_{\mathbb{V}(\mathcal{T})}$ is the Higgs branch.

Fixed components vs simple modules

- For a 4-punctured sphere, VOA is affine D_4 Lie algebra $\mathbb{V}(\mathcal{T}_{0,4}) = L_{-2}(\mathfrak{d}_4)$ with level -2
- There are 5 simple modules:

$$0, -2\omega_1, -2\omega_3, -2\omega_4, -\omega_2$$

- Fixed components under the $U(1)_r$ action consists of 4 points and $\text{Bun}_G \cong \mathbb{P}^1$:



Fixed components vs simple modules

- The $\mathbb{V}(\mathcal{T}_{0,4})$ -module characters $\{\chi_\lambda\}$ satisfy a set of flavor modular differential equations (FMDEs) These form a set of partial differential equations.
- The quasi-modularity of FMDEs implies that characters belong to a representation $\mathbb{S}_{0,4}^{mod}$ of the modular group
- Generally, the dimension of $\mathbb{S}_{0,4}^{mod}$ is larger than the number of SMs, where the extra solutions are usually logarithmic modules, and T is not semisimple.

$$SL(2, \mathbb{Z}) \curvearrowright \mathbb{S}_{0,4}^{mod} = \text{Span}_{\mathbb{C}} \{ 5 \text{ characters} + \log\text{-module} \} , \quad \text{Jord}(T) = [2, 1^4]$$

- Another perspective comes from the mixed Hodge polynomial of character variety We need its pure part:

$$\text{PH}(\mathcal{M}_{0,4}^{\text{Betti}}; q) = 4 + q$$

- $\#$ of simple modules = $\#$ fixed manifolds = $\#$ Jordan block $(T) = PH(q=1)$

Fixed components vs simple modules

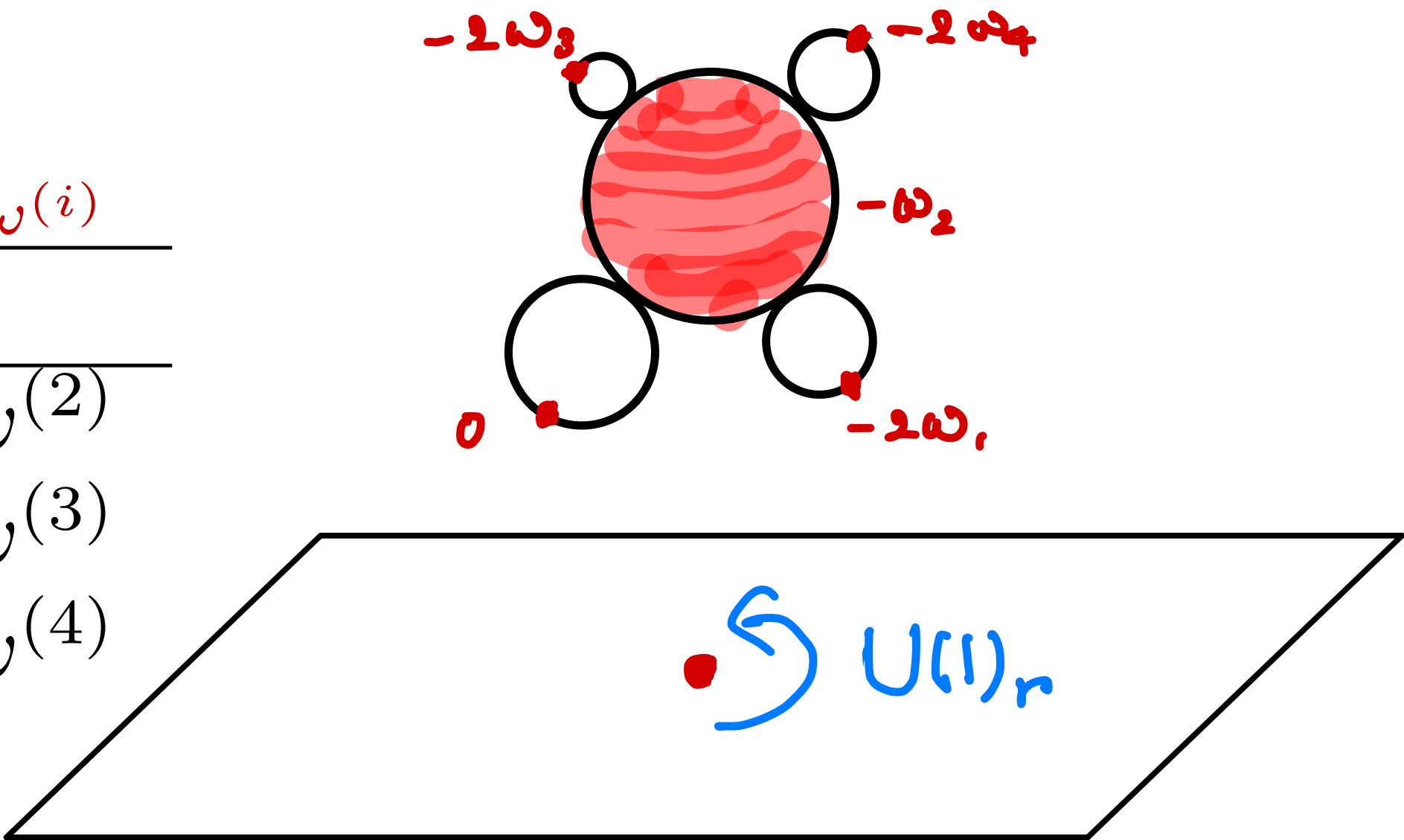
Yiwen Pan-Wenbin Yan

- $SL(2, \mathbb{Z}) \curvearrowright \mathbb{S}_{0,4}^{mod} = \text{Span}_{\mathbb{C}} \{ 5\text{characters} + \text{log-module} \} , \quad \text{Jord}(T) = [2, 1^4]$
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$$\text{PH}(\mathcal{M}_{0,4}^{\text{Betti}}; q) = 4 + q$$

- # of simple modules = # fixed manifolds = # Jordan block $(T) = PH(q = 1)$

$L(\omega_i)$	h	$\mathfrak{sl}(2)^{\otimes 4}$ rep	$\dim M$	$\mu - \mu_{max} _{\alpha_i \mapsto -\omega^{(i)}}$
$-\omega_2$	-1	$[-2, 0, 0, 0]$	1	$-1 - 2\omega^{(1)}$
$-2\omega_1$	-1	$[-2, -2, 0, 0]$	0	$-1 - 2\omega^{(1)} - 2\omega^{(2)}$
$-2\omega_3$	-1	$[-2, 0, -2, 0]$	0	$-1 - 2\omega^{(1)} - 2\omega^{(3)}$
$-2\omega_4$	-1	$[-2, 0, 0, -2]$	0	$-1 - 2\omega^{(1)} - 2\omega^{(4)}$
$-2\omega_0$	0	$[0, 0, 0, 0]$	0	0



SECTION 3

Time for wine? I bought a new decanter!

1-punctured torus

- Moduli space of flat $\mathrm{SL}(2, \mathbb{C})$ -connections on a once-punctured torus C_p

$$\pi_1(C_p) = \langle \mathfrak{m}, \mathfrak{l}, \mathfrak{c} | \mathfrak{m} \mathfrak{l} \mathfrak{m}^{-1} \mathfrak{l}^{-1} = \mathfrak{c} \rangle .$$

- These variables are subject to the cubic equation

$$\mathcal{M}_{\mathrm{flat}}(C_p, \mathrm{SL}(2, \mathbb{C})) = \{(x, y, z) \in \mathbb{C}^3 | x^2 + y^2 + z^2 - xyz - 2 = t^2 + t^{-2}\} .$$

From the 4-punctured sphere, we can obtain it by setting $\theta_{1,2,3} = 0$.

- The deformation quantization of $\mathcal{O}(\mathfrak{X})$ with respect to Ω_J is the spherical DAHA $S\ddot{H}_{q,t}$ of type A_1

$$\mathcal{O}(\mathfrak{X}) \rightsquigarrow \mathcal{O}^q(\mathfrak{X}) \cong S\ddot{H}_{q,t}$$

- Relations are given by:

$$[x, y]_q = (q^{-1} - q)z$$

$$[y, z]_q = (q^{-1} - q)x$$

$$[z, x]_q = (q^{-1} - q)y$$

$$q^{-1}x^2 + qy^2 + q^{-1}z^2 - q^{-\frac{1}{2}}xyz = (q^{-\frac{1}{2}}t - q^{\frac{1}{2}}t^{-1})^2 + (q^{\frac{1}{2}} + q^{-\frac{1}{2}})^2 ,$$



Symmetry

- **Modular group** $\mathrm{PSL}(2, \mathbb{Z})$: Since the algebra is constructed from a torus, it enjoys $\mathrm{PSL}(2, \mathbb{Z})$ -automorphism generated by

$$\tau_+ \mapsto \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad \tau_- \mapsto \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

and

$$\sigma := \tau_+ \tau_-^{-1} \tau_+ = \tau_-^{-1} \tau_+ \tau_-^{-1} \mapsto \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \tau_+ \tau_-^{-1} \mapsto \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}.$$

The actions on the generators are given by

$$\tau_+ : (x, y, z) \mapsto \left(x, \frac{xy + yx}{q^{1/2} + q^{-1/2}} - z, y \right),$$

$$\tau_- : (x, y, z) \mapsto \left(\frac{xy + yx}{q^{1/2} + q^{-1/2}} - z, y, x \right),$$

$$\sigma : (x, y, z) \mapsto \left(y, x, \frac{xy + yx}{q^{1/2} + q^{-1/2}} - z \right).$$

- “**sign changes**” $\Xi = \mathbb{Z}_2 \times \mathbb{Z}_2$ are holomorphic involutions, and its generators ξ_1, ξ_2 and their combination $\xi_3 := \xi_1 \circ \xi_2$ act as

$$\xi_1 : (x, y, z) \mapsto (-x, y, -z),$$

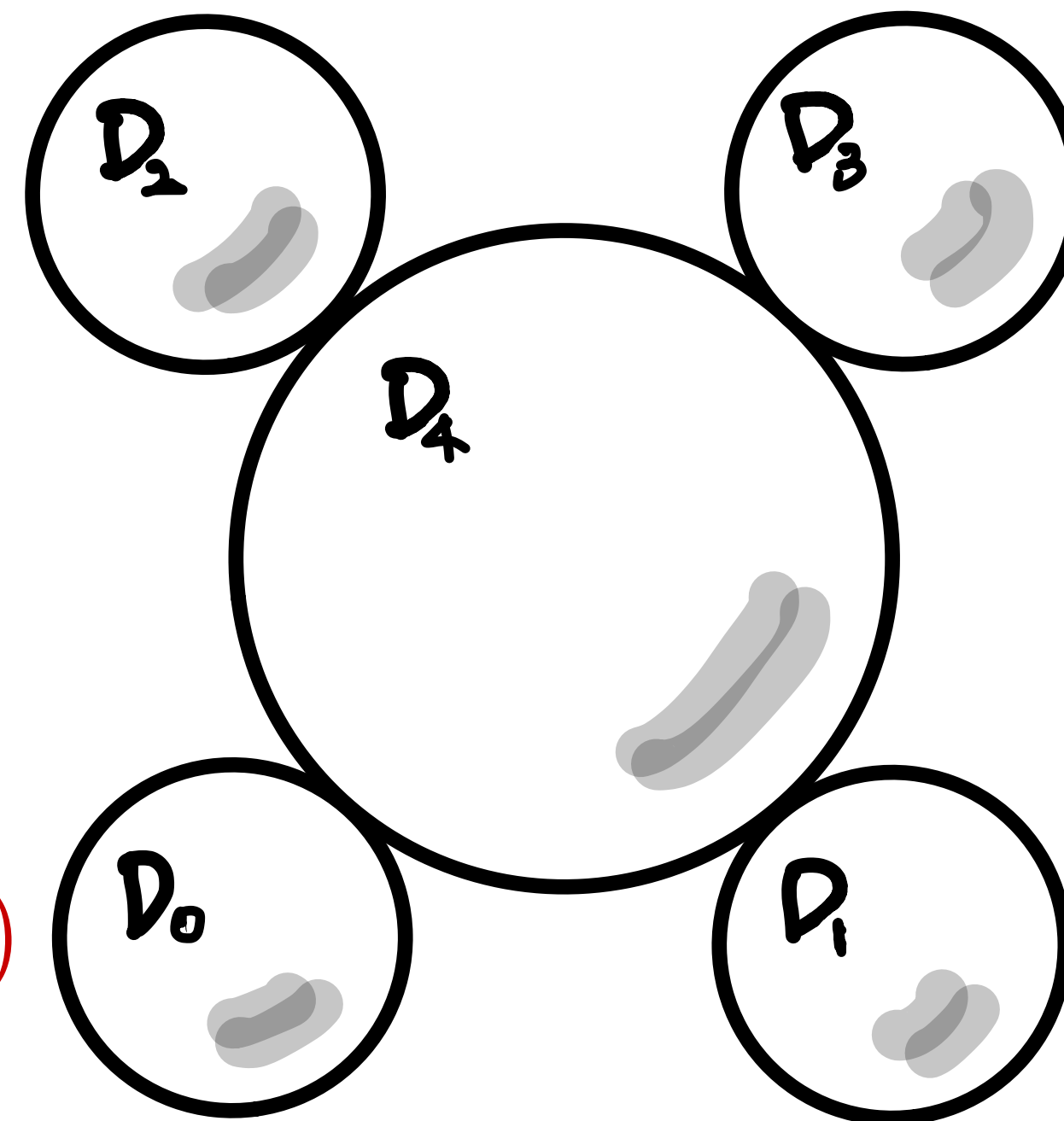
$$\xi_2 : (x, y, z) \mapsto (x, -y, -z),$$

$$\xi_3 : (x, y, z) \mapsto (-x, -y, z),$$



Modularity

$PSL(2, \mathbb{Z})$ -actions on $\mathfrak{B}_{D_0}$ and $\mathfrak{B}_{D_4}$ are categorified by modular tensor categories



Affine Lie algebra $L_k(\mathfrak{a}_1)$

$PSL(2, \mathbb{Z}) \curvearrowright \text{Hom}(\mathfrak{B}_{D_4}, \mathfrak{B}_{cc})$

Virasoro minimal model

$PSL(2, \mathbb{Z}) \curvearrowright \text{Hom}(\mathfrak{B}_{D_0}, \mathfrak{B}_{cc})$

$(2, 2\ell + 1)$ Virasoro minimal model is VOA of $(A_1, A_{2(\ell-1)})$ Argyres-Douglas theory

(A_1, A_2) Argyres-Douglas theory and Lee-Yang model

$$\text{circle with square} \quad \phi(z) \sim \frac{\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}}{z^{2+\frac{3}{2}}} + \dots$$

$$\mathbb{V}(A_1, A_2) = \text{Lee-Yang minimal model}$$

$$2 \text{ primaries } \{\varphi_0, \varphi_{-\frac{2}{5}}\} \curvearrowright PSL(2, \mathbb{Z})$$

$$S = \frac{2}{\sqrt{5}} \begin{pmatrix} -\sin \frac{2\pi}{5} & \sin \frac{\pi}{5} \\ \sin \frac{\pi}{5} & \sin \frac{2\pi}{5} \end{pmatrix}, \quad T = \begin{pmatrix} e^{\frac{11\pi i}{30}} & 0 \\ 0 & e^{-\frac{\pi i}{30}} \end{pmatrix}$$

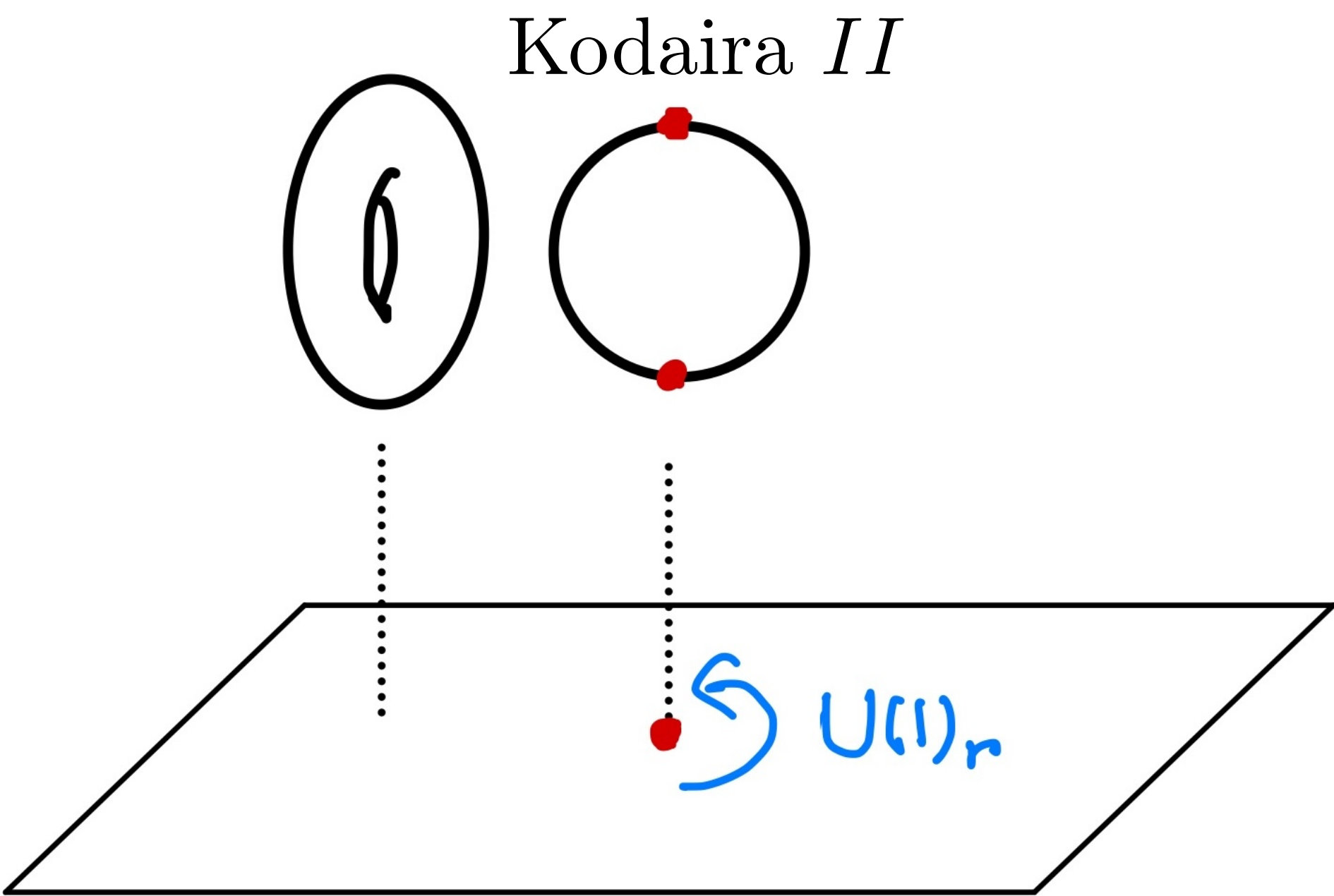
Bezrukavnikov-Boixeda Alvarez-McBreen-Yun

2 fixed points in the central fiber (Kodaira type II)

In wild Hitchin moduli space

the central fiber is affine Springer fiber

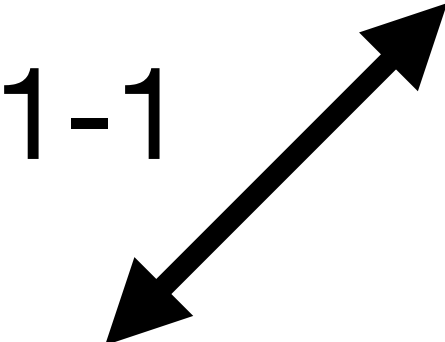
$$\mathbb{C}^\times \curvearrowright Fl_\gamma = \{g \in G((z))/G[[z]] \mid Ad_{g^{-1}}(\gamma) \in \mathfrak{g}[[z]]\}$$



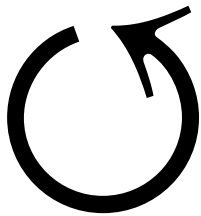
Web of relations

Simple modules of affine Lie algebras
with boundary admissible level

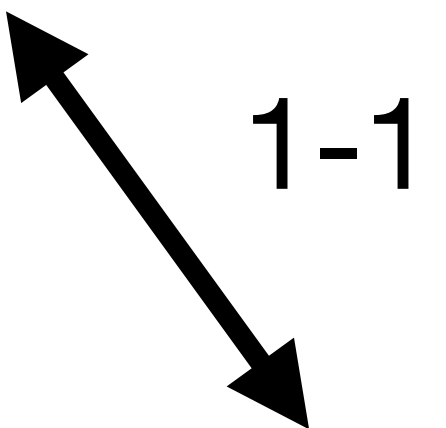
Peng Shan-Wenbin Yan-Dan Xie



$\mathbb{C}^\times$ -fixed points of
Affine springer fiber

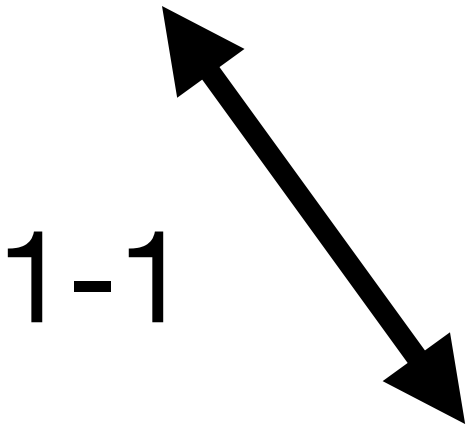


$PSL(2, \mathbb{Z})$



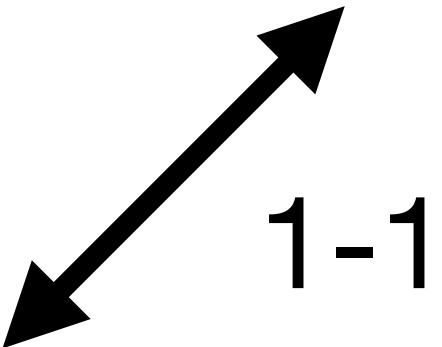
Finite-dimensional module
of DAHA

$$t^{h^\vee} q^n = 1$$
$$q^m = 1$$



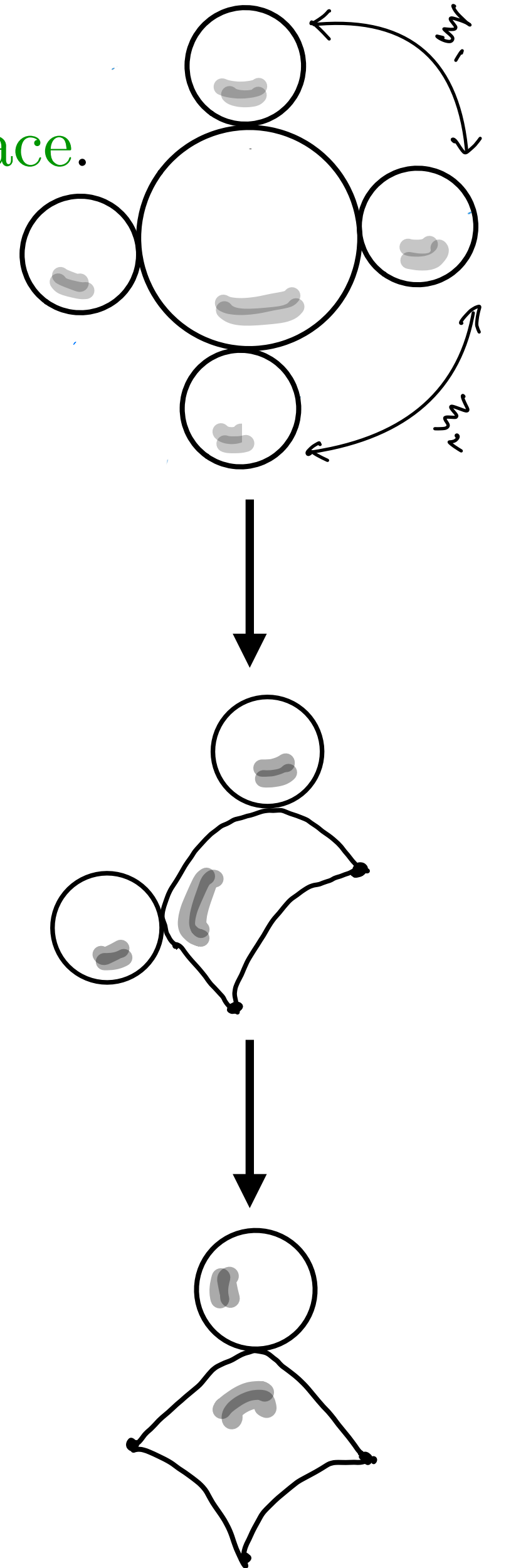
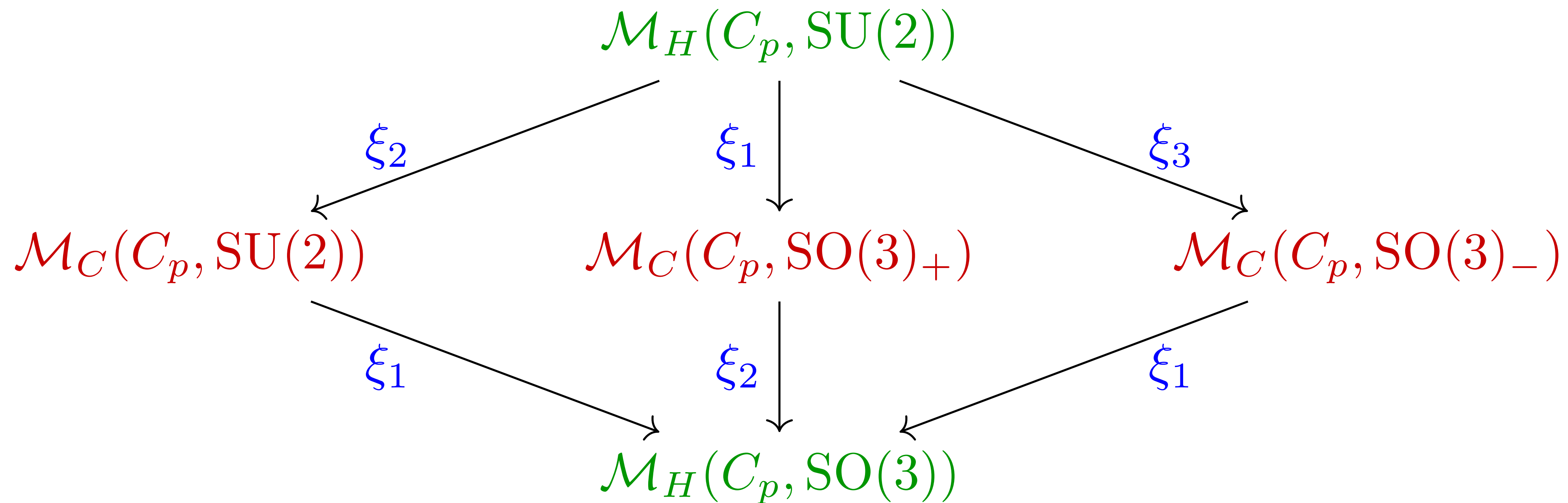
Bezrukavnikov-Boixeda Alvarez-McBreen-Yun

$\mathbb{C}^\times$ -fixed points of wild Hitchin moduli space



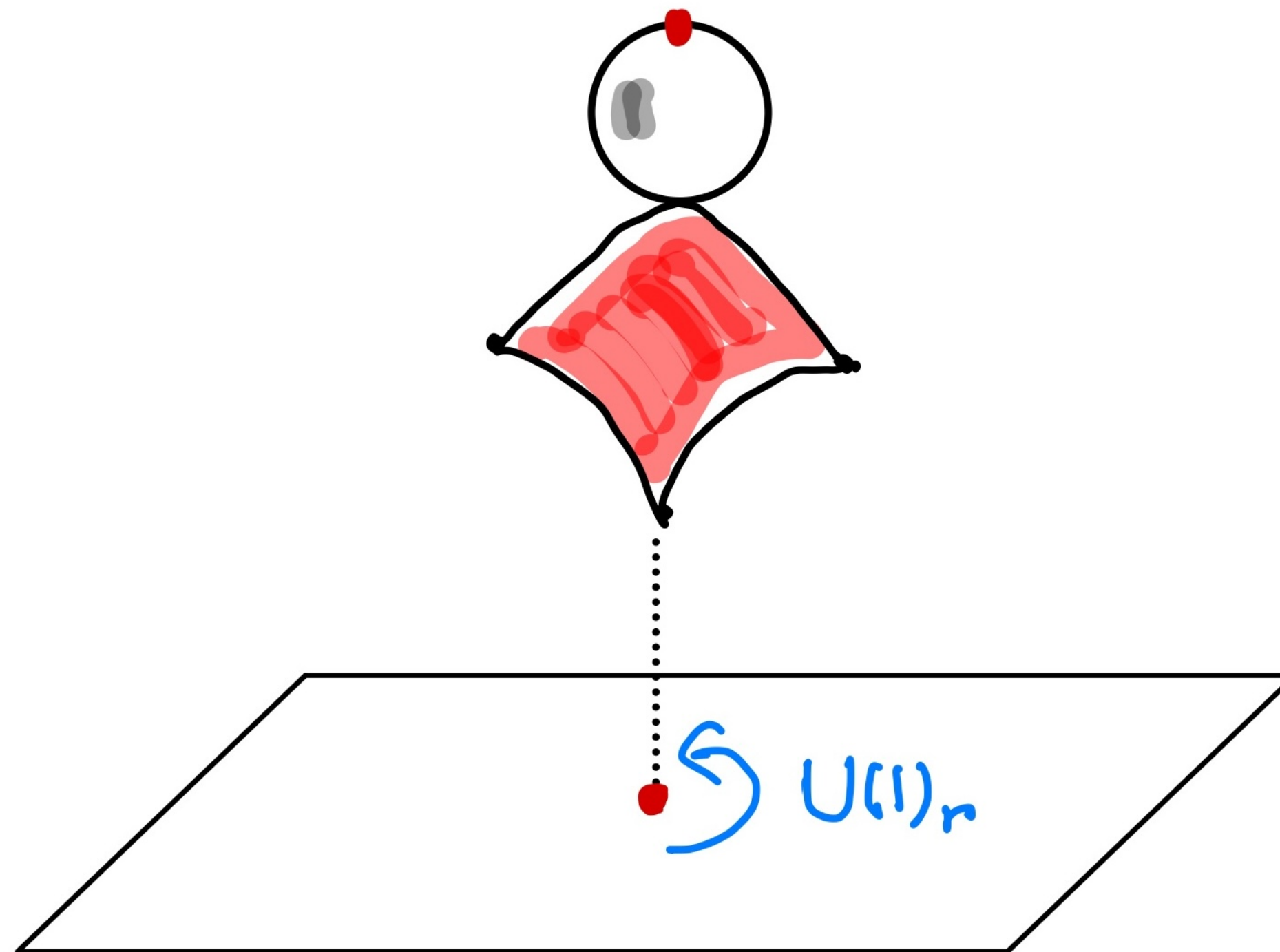
Hitchin moduli space and Coulomb branch

Coulomb branch of 4d $\mathcal{N} = 4$ theory is a $\mathbb{Z}_2$ quotient of $SU(2)$ Hitchin moduli space.



Simple modules vs fixed points via Langlands duality

- For a **once-punctured torus**, **VOA** $\mathbb{V}(\mathcal{T}_{0,1})$ is **small** $\mathcal{N} = 4$ **superconformal Virasoro algebra** with $c = -9$.
- There are **2 simple modules** of $\mathbb{V}(\mathcal{T}_{0,1})$: $0, -\omega_1$.
- $\mathbb{S}_{1,1}^{\text{mod}} = \text{Span}_{\mathbb{C}}(\text{2 simple} + \text{log})$, **Jordan type** of T is $[2,1]$
- **Fixed manifolds** of **SO(3) Hitchin moduli space** $\mathcal{M}_H(C_p, \text{SO}(3))$ consist of a point $+ \mathbb{P}^1 / (\mathbb{Z}_2 \times \mathbb{Z}_2)$
- $\text{SU}(2) \leftrightarrow \text{SO}(3)$ Langlands duality **Gukov-Pei-Hsin**



Mysterious relationship

$$\text{Simple modules of } \mathbb{V}(\mathcal{T}(C_{g,n}, G)) \overset{1-1}{\longleftrightarrow} \mathbb{C}^\times\text{-fixed components of } \mathcal{M}_H(C_{g,n}, G^\vee)$$

- More precise correspondence has been formulated for $G=\text{SU}(2)$ Yiwen Pan-Wenbin Yan
- Very little is understood for higher-rank
- Koszul type duality? What is the underlying intrinsic reason?
- Very mysterious even for physicists
- Any insights from mathematicians will be appreciated